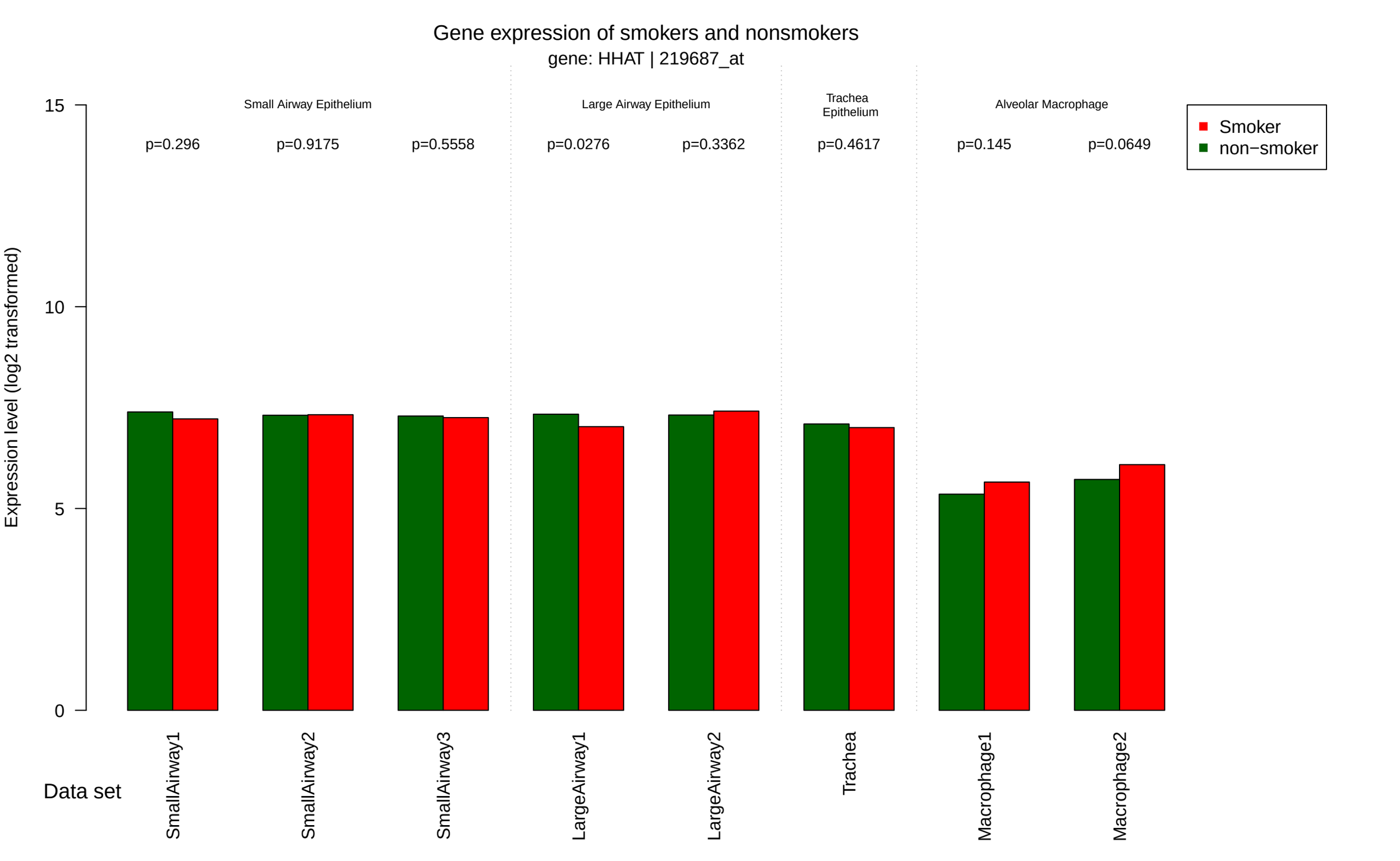
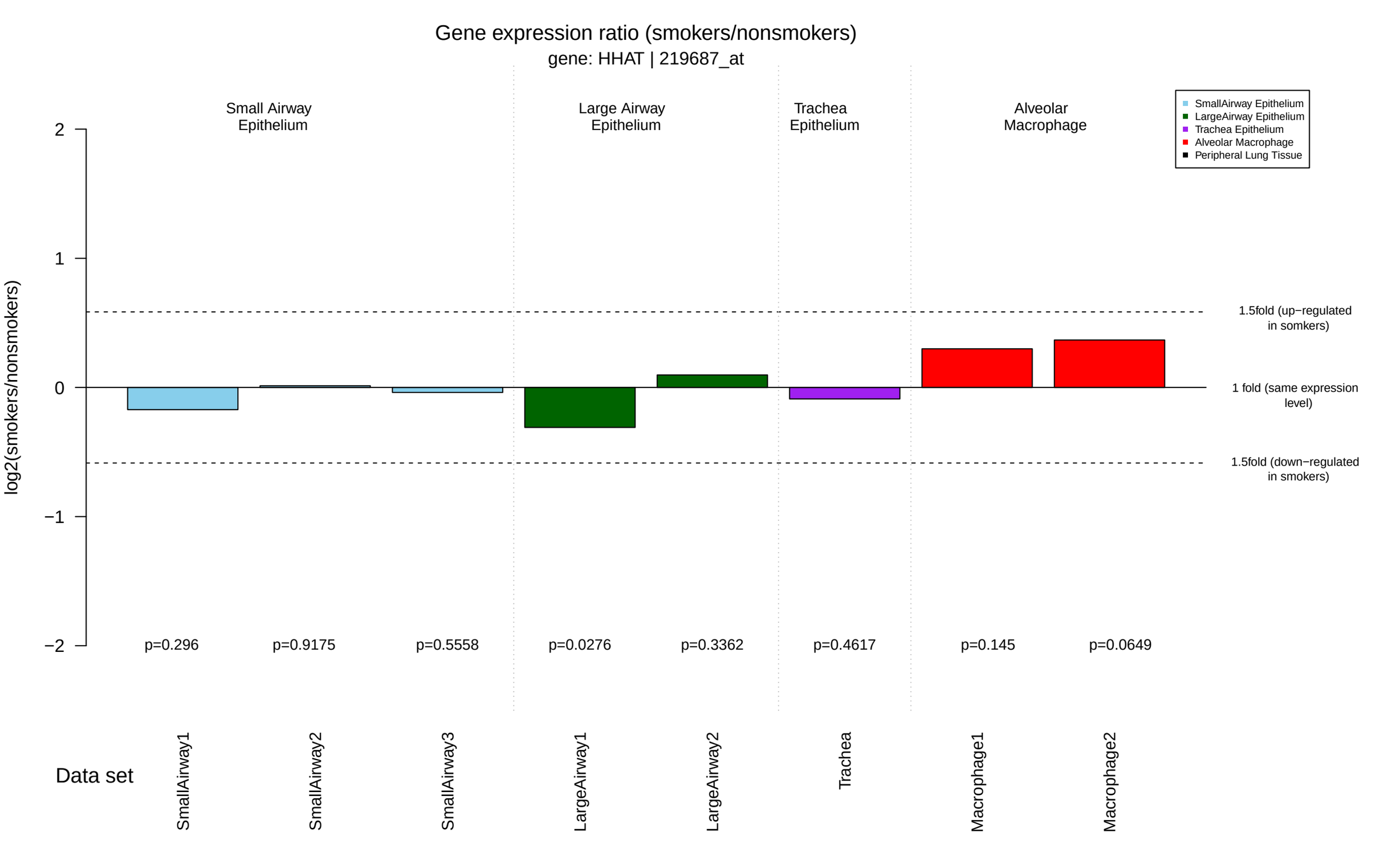
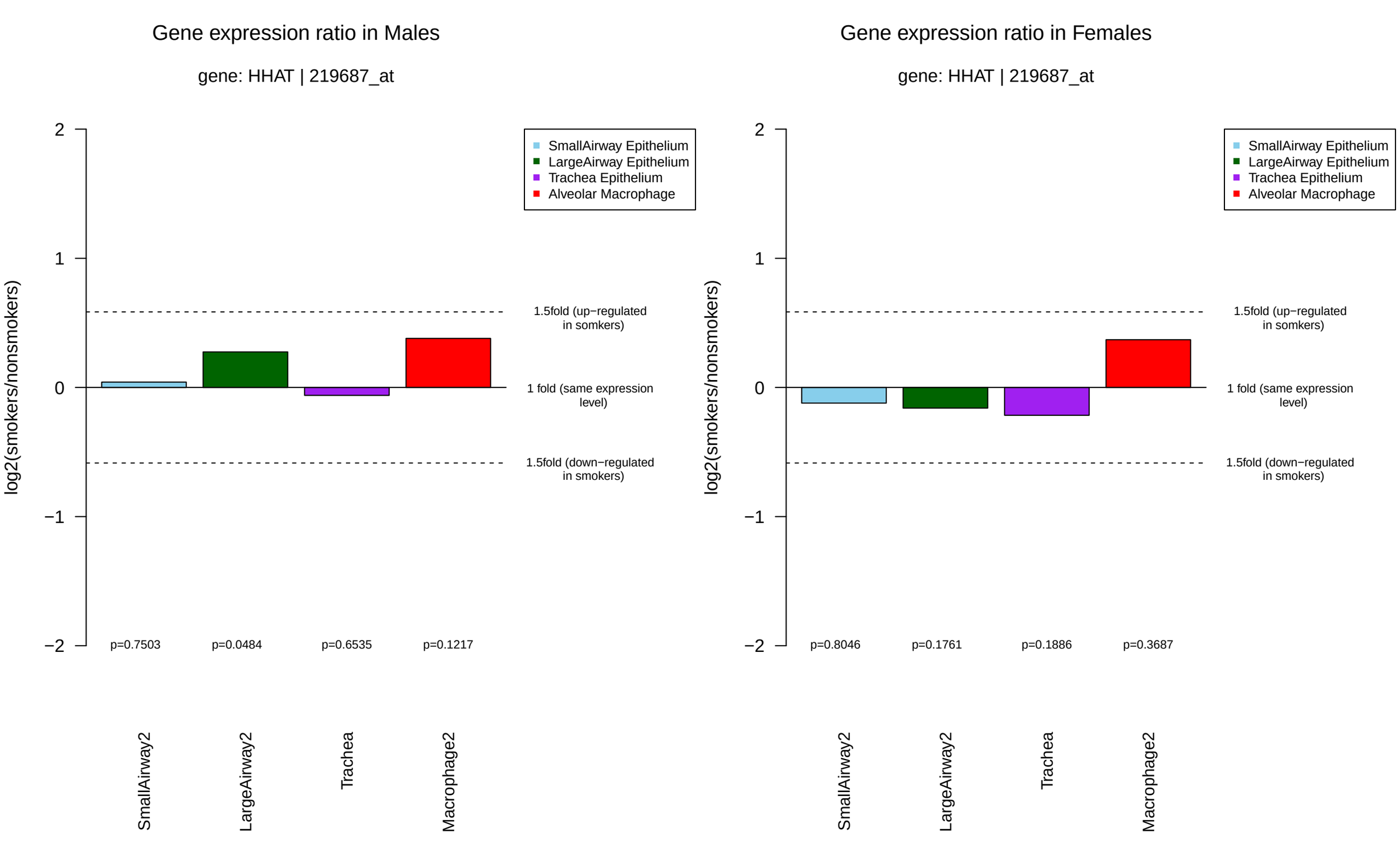
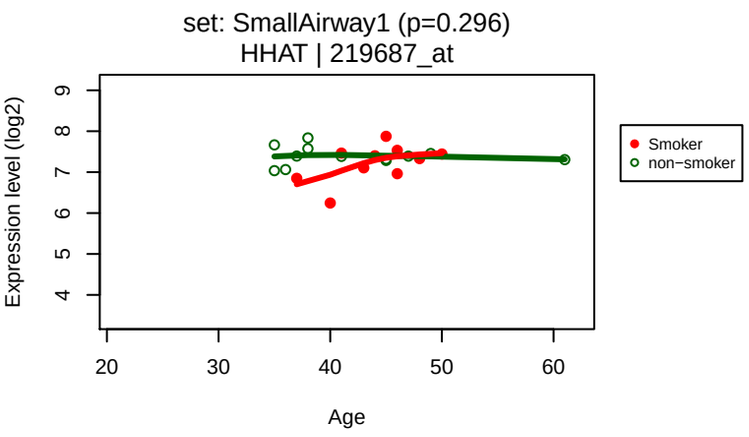
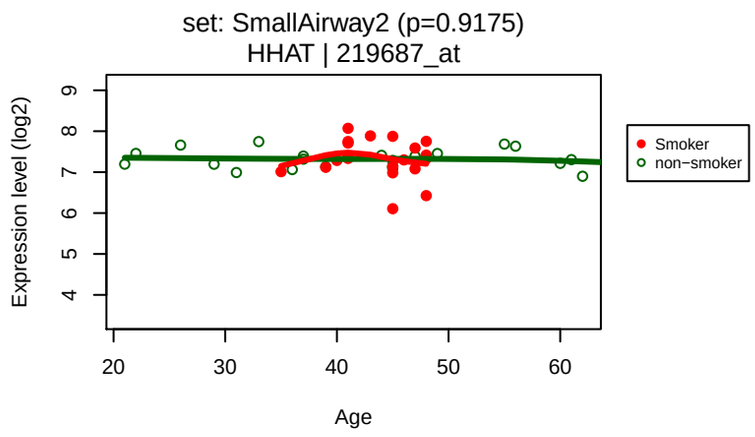
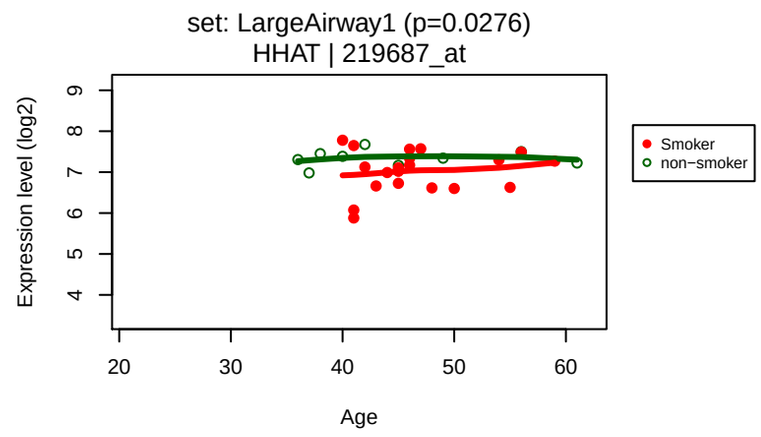

Gene expression of smokers and nonsmokers
gene: HHAT | 219687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.296
p=0.9175
p=0.5558
p=0.0276
p=0.3362
p=0.4617
p=0.145
p=0.0649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

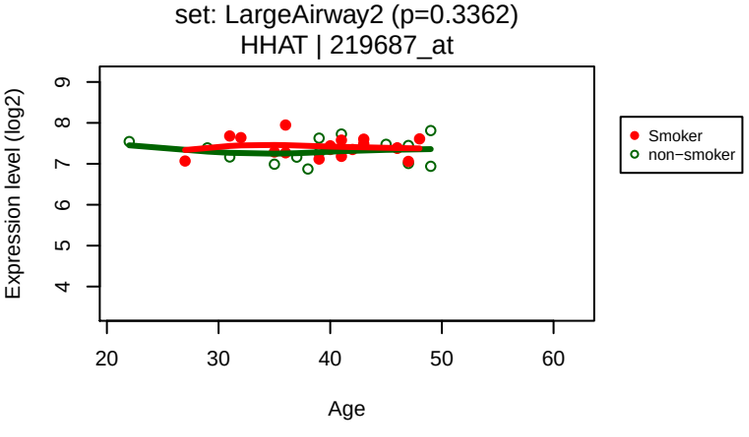

Gene expression ratio (smokers/nonsmokers)
gene: HHAT | 219687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.296
p=0.9175
p=0.5558
p=0.0276
p=0.3362
p=0.4617
p=0.145
p=0.0649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

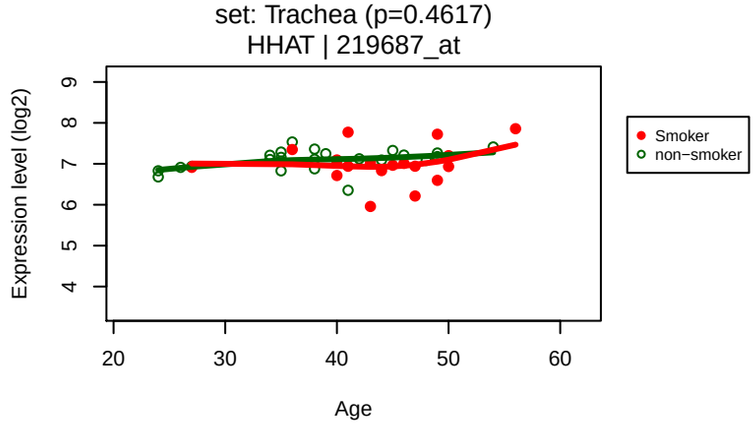

Gene expression ratio in Males
gene: HHAT | 219687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7503
p=0.0484
p=0.6535
p=0.1217
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HHAT | 219687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8046
p=0.1761
p=0.1886
p=0.3687
SmallAirway2
LargeAirway2
Trachea
Macrophage2

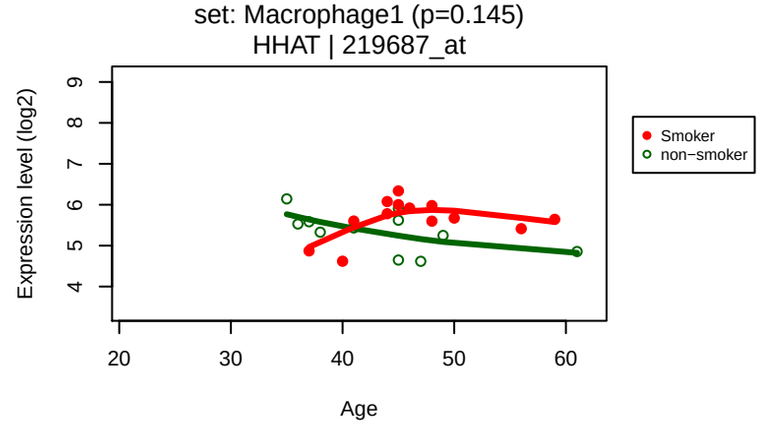

set: SmallAirway1 (p=0.296)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker

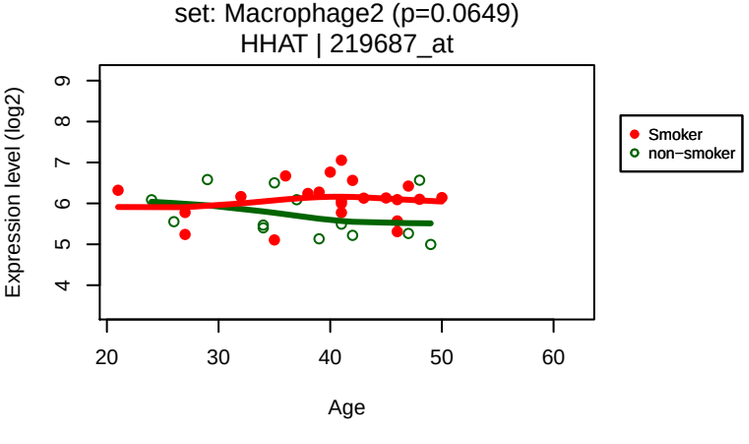

set: SmallAirway2 (p=0.9175)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker

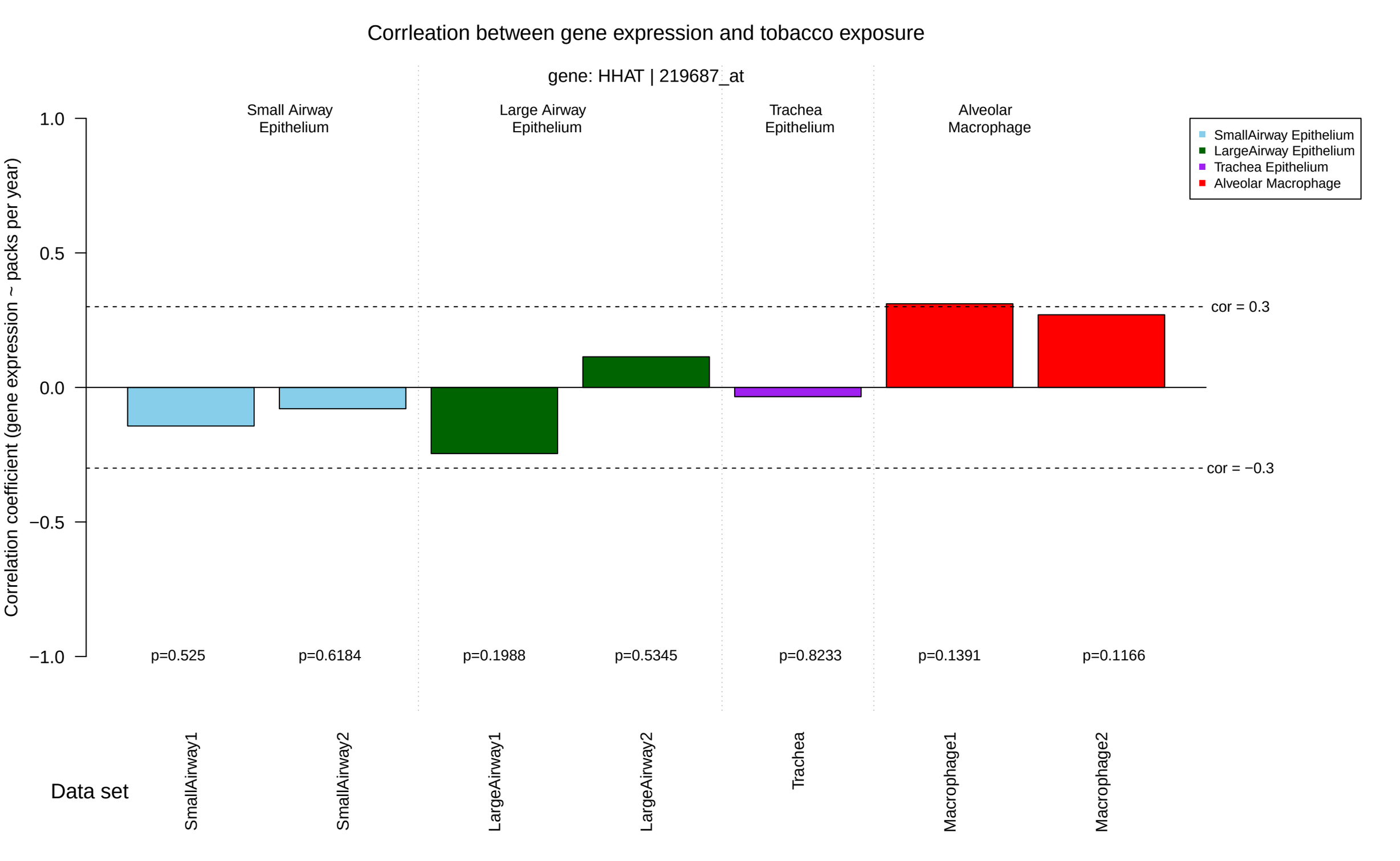

set: LargeAirway1 (p=0.0276)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3362)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4617)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.145)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0649)
HHAT | 219687_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HHAT | 219687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.525
p=0.6184
p=0.1988
p=0.5345
p=0.8233
p=0.1391
p=0.1166
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2